**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 216 · Quicksort: pivot and partition

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. In bubble sort, which value reaches its final position at the end of pass one?

Q2. What is a base case?

Q3. Which element is a[3] in [6,2,9,4,5]?

Q4. What is the result of 4 <= 6 and why?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

pivot — Element selected to split a quicksort partition.

partition — Rearrange values so smaller/equal values are on one side of the pivot.

Lomuto — A partition approach scanning left to right and placing the pivot last.

index — Zero-based position within the array.

divide and conquer — Solve smaller subproblems and combine results.

## New learning

Pick a pivot — Use the rightmost element, 6. A left boundary i tracks where the next small value belongs.

Scan and move — Only values less than or equal to 6 move left. We move 2, then 4, without sorting those two values.

Place the pivot — Swap 6 into index 2; elements to the left are <=6, and those to the right are >6. Each side still needs sorting.

## Worked example

Partition [6,3,8,2,5] around final-element pivot 5.

1. Predict the next state — 6 is larger than 5; 3 is smaller and swaps into the left boundary.

2. Reveal and explain — 8 is larger; 2 swaps to index 1, giving [3,2,8,6,5].

3. Reveal and explain — Swap pivot 5 with position 2: [3,2,5,6,8]. The two sides are not yet necessarily sorted.

# We Do · Practise with guidance

Partition [8,1,7,3,6] using final pivot 6.

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Record every move of a value <= 6.
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Predict the next state

Compare 8 and 1 with pivot 6. Move 1 into index 0.

Reveal and explain

7 is too large, then 3 moves to index 1: [1,3,7,8,6].

Put pivot at index 2: [1,3,6,8,7]. The right side [8,7] still needs sorting.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Partition [4,1,3] using the final value as pivot, with Lomuto partition. Give the result and pivot index.

### Question 2

Partition [9,2,5,1,6] using the final pivot. Show the array after moving each qualifying value and after placing the pivot.

### Question 3

Explain why choosing the last element as pivot can lead to $O(n^2)$ work on already sorted data, and suggest an improvement.

# Exit Ticket · Answer before checking

A quicksort partition automatically fully sorts the two sides of the pivot.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. The largest value in the unsorted section.

Q2. A condition that stops recursive calls.

Q3. 4; indexes begin at 0.

Q4. True, because 4 is less than or equal to 6.

### You Do pages · Independent questions

Q1. [3] Pivot=3 [1]; move 1 left [1]; result [1,3,4] with pivot index 1 [1].

Q2. [5] Move 2: [2,9,5,1,6] [1]; move 5: [2,5,9,1,6] [1]; move 1: [2,5,1,9,6] [1]; pivot swap: [2,5,1,6,9] [1]; pivot index 3 [1].

Q3. [4] An already sorted input can place each pivot at one extreme [1]; partitions become n−1 and 0 elements [1]; recursion depth and comparisons become quadratic [1]; randomised/median-of-three pivot selection can reduce repeated poor pivots in practice [1].

### Exit Ticket · Model answer

False. Partition places the pivot correctly; the two sides require further sorting.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding